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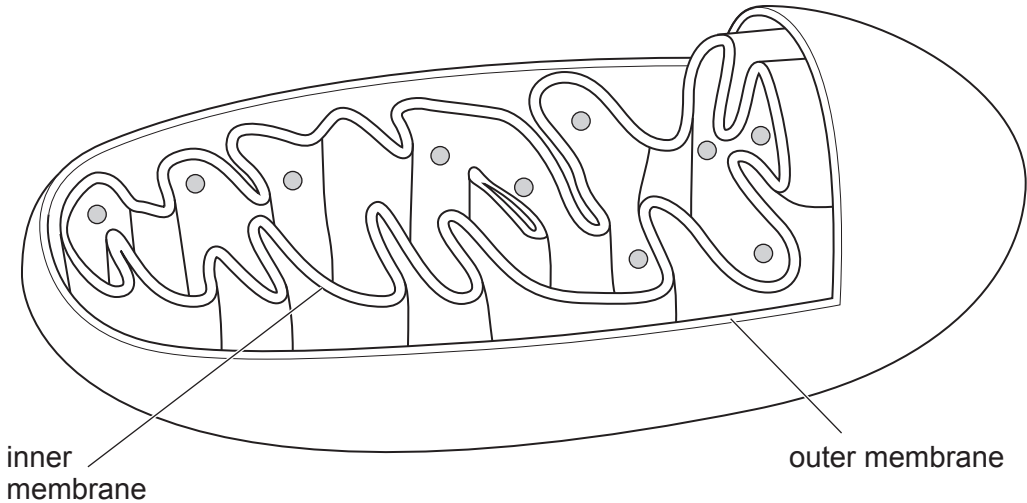


Fig. 8.1

Outline the roles played by the mitochondrial membranes in respiration.

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(b) The shapes and numbers of mitochondria are continually changing due to fission. Fission occurs when one mitochondrion splits to form two mitochondria.

(i) Suggest reasons why mitochondria carry out fission.

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(ii) Guanosine triphosphate (GTP) is a molecule that is used as a source of energy in some reactions, instead of ATP. Guanosine is composed of a purine, similar to adenine, and ribose.

Suggest why GTP can be a suitable source of energy in some reactions.

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(c) Old or damaged mitochondria reduce the ability of a cell to carry out aerobic respiration and produce the ATP needed for the metabolic processes of the cell.

Suggest what occurs to these mitochondria to allow the cell to maintain the same overall rate of respiration and ATP production.

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[Total: 9]